

Eric (Chen-Chien) Chang

BACKEND / DATA INFRASTRUCTURE ENGINEER

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PROFILE

Backend and data infrastructure engineer building real-time data products end to end. Experienced in distributed systems, performance engineering, MCP, and open-source data platforms.

EXPERIENCE

Lawsnote

Jan 2024 – Dec 2024

Taipei, Taiwan

Backend Software Engineer

- Implemented **HTTP streaming** for CPU-intensive computation results, reducing first-byte latency from **100s to 5s**.
- Built **MinIO event** and **Elasticsearch** monitoring to detect missing data and improve pipeline reliability.
- Profiled and fixed a **NAPI-rs** performance issue with **Pyroscope**, improving Node.js native package execution.
- Built an **LLM service** for regulatory text extraction and automated government-act crawl and re-parse workflows.

MediaTek (Contract)

Feb 2022 – Mar 2023

Hsinchu, Taiwan

Backend Software Engineer

- Cut data-warehouse WebUI load time from **60s to 1s** with **PostgreSQL index-only scans** (case study).
- Cut simulation jobs from **2hr to 1min** by profiling with **Pyroscope eBPF** and applying PostgreSQL bulk import.
- Designed a queue-based microservice architecture and operated a **5-node** cluster for long-running jobs.
- Cut Go service memory use by **50%** with **sync.Pool**; instrumented it with Prometheus, OpenTelemetry, and Pyroscope.

National Cheng Kung University

Sep 2017 – Jul 2019

Tainan, Taiwan

Research Assistant

- Conducted FDM 3D-printing research and built experimental tooling with Python, C, MicroPython, and Arduino.

SELECTED PRODUCT

Locus — Real-World Context Layer for AI Assistants

2026 – Present

Cloudflare Workers / R2 / MCP

Independent Product

- Built and deployed a remote **Streamable HTTP MCP** data product that turns natural-language destinations into ranked parking decisions across Taipei, Hsinchu, and Zhubei.
- Designed a five-minute serverless ingestion and materialization pipeline, normalizing heterogeneous municipal and TDX feeds into one provenance-aware contract for **1,271 facilities** (1,238 live-count, 33 static-only).
- Implemented OSM geocoding, regional validation, freshness and coverage semantics, and availability and straight-line distance ranking with Google Maps handoff; passed all six cross-region source-consistency validations against independently fetched authoritative data.

OPEN SOURCE LEADERSHIP & CONTRIBUTIONS

- **Apache Kafka Contributor** — collaborating on Kafka Streams **KIP-770** input-buffer byte bounds (PR #22458); fixed JVM container awareness and offset commit-before-partition-switch correctness (PR #21584).
- **Apache Gravitino Committer** — added PostgreSQL/MySQL integration-test container management and reproducible Docker image workflows; reduced the Hive image by **420MB** and ported Python client partition APIs.
- **Apache DataFusion / Comet Contributor** — implemented Spark-compatible `str_to_map` in Rust (PR #20120) and wired Comet's native SparkStrToMap UDF with SQL coverage (PR #3654).
- **redis/rueidis Contributor** — implemented the go-redis API Adapter for teams migrating from go-redis (PR #395).

TECHNICAL SKILLS

Languages: Go, Java, Rust, JavaScript / TypeScript, Python, C

Data & processing: PostgreSQL, ClickHouse, Hive, Kafka, Kafka Streams, Flink, Spark, SQL

Infrastructure: Airflow, Docker, Kubernetes, Linux, GitHub Actions

Observability: Prometheus, Grafana, OpenTelemetry, Pyroscope, Great Expectations

EDUCATION

National Taiwan Ocean University — Electrical Engineering coursework (Drop Out)

2013 – 2017